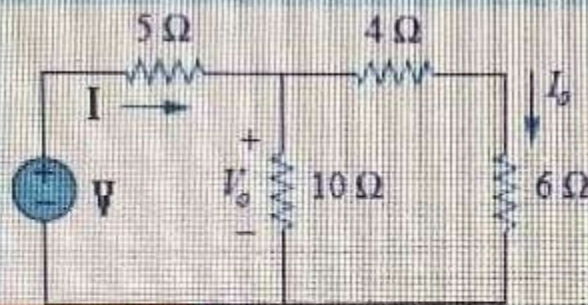


Consider the circuit shown below. When the voltage $V_o = 4V$, then the value of I is



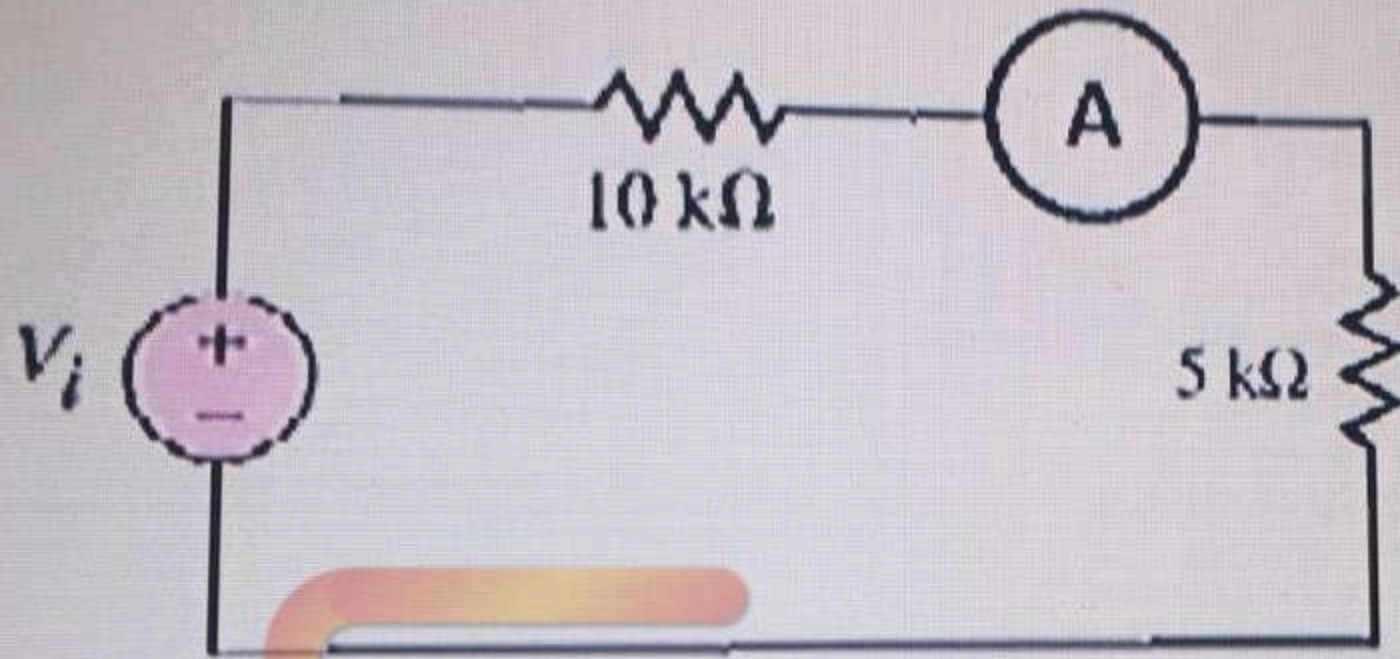
Select one:

☒ a. 0.8 A

☐ b. 0.4 A

☐ c. 1.2 A

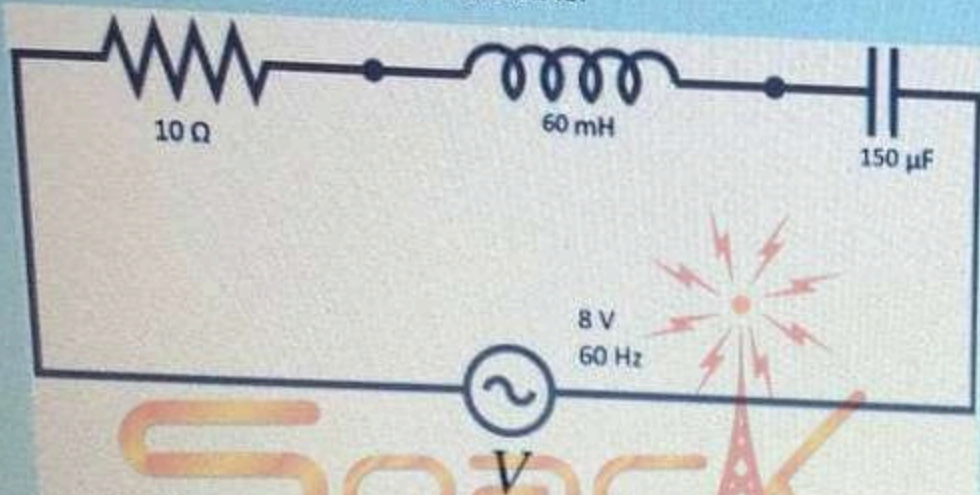
For the circuit shown below, when the voltage source is a 60 peak to peak voltage, the ammeter reading is:



Select one:

- ☐ A. 1.76 mA
- ☐ B. 1.91 mA
- ☐ C. 1.04 mA
- ☒ D. 1.41 mA

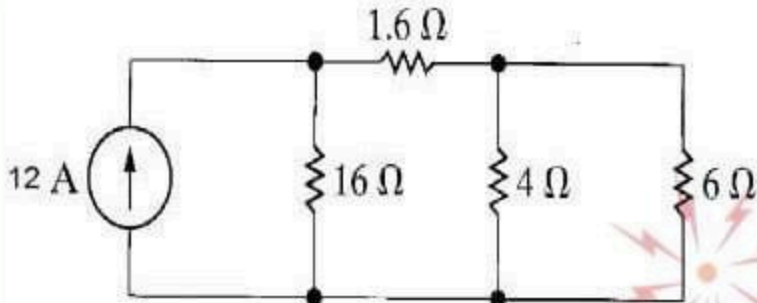
The resonance frequency in this circuit is:



Select one:

- ☒ A. 58.3 Hz
- ☐ B. 53 Hz
- ☐ C. 41.1 Hz
- ☐ D. 64 Hz

In the circuit shown below, the current
:through (4 ohm) resistor is



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A. 4.8 A ☐

B. 9.6 A ☐

C. 3.84 A ☐

D. 3 A ☐

5.76

Voltmeter in AC circuits reads the voltage as

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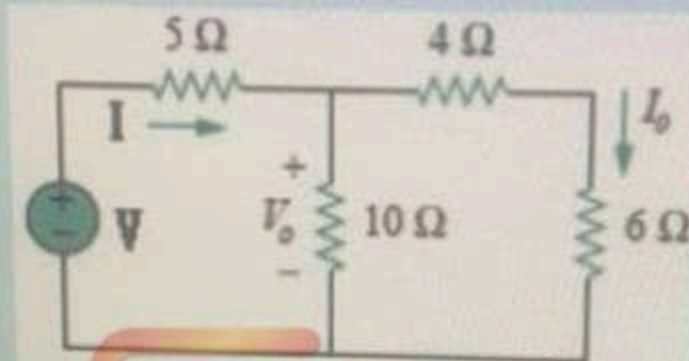
a. RMS voltage ☒

b. Peak to peak voltage ☐

c. Transient voltage ☐

d. Peak voltage ☐

Consider the circuit shown below. The value of the current I_o , when $I = 0.6$ A, is



Select one:

☒ a. 0.3 A

☐ b. 0.4 A

☐ c. 0.2 A

If the voltage source is 30 volt, then the current through impedance ($Z=6+j12$) is:

Select one:

☐ A. 3.74 A

☐ B. 3.12 A

☐ C. 1.63 A

☒ D. 2.23 A

:This impedance ($Z = 5 - j11$) represent

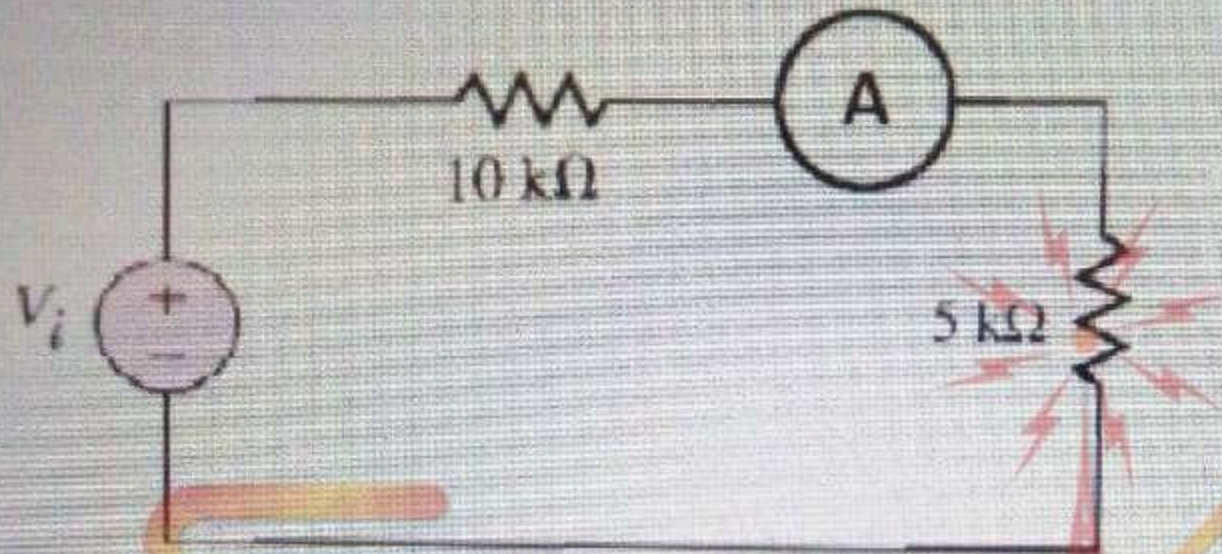
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A. Inductive load ☐

B. Resistive load ☐

C. Capacitive load ☒

For the circuit shown below, when the voltage source is a 75 peak to peak voltage, the ammeter reading is:



Select one:

- ☐ A. 1.04mA
- ☐ B. 1.91mA
- ☒ C. 1.76 mA
- ☐ D. 1.41 mA

If the voltage source is 40 volt, then the current
:through impedance ($Z=8+j20$) is

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A. 3.12 A ☐

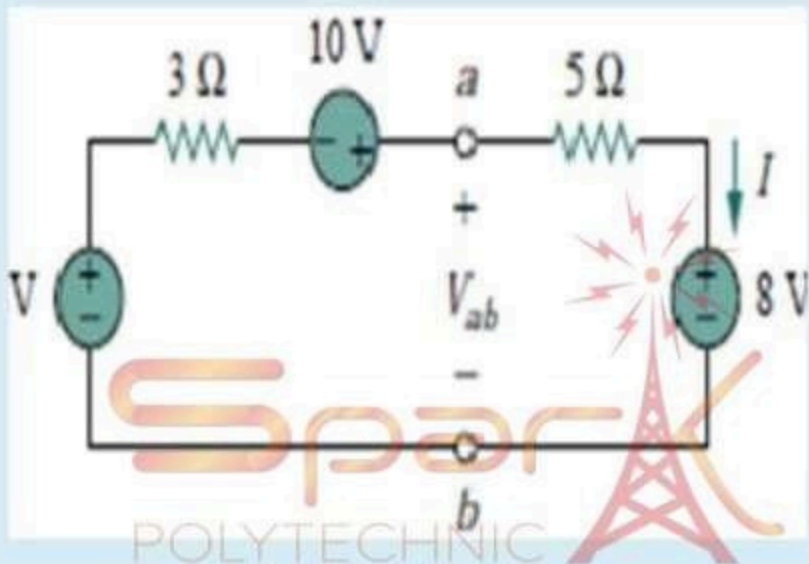
B. 2.23 A ☐

C. 1.86 A ☒

D. 3.74 A ☐

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Consider the circuit shown below, when $I = 0.75 \text{ A}$ then the value of the voltage V



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a. 6 V ☐

b. 8 V ☐

c. 4 V ☒

Below the resonant frequency,
the load will become

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A. Resistive ☐

B. Capacitive ☒

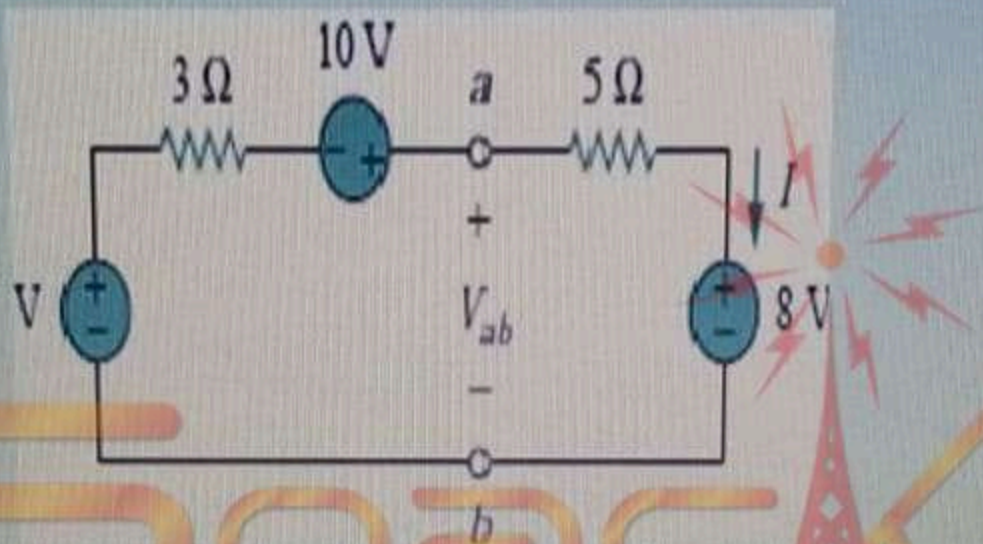
C. Inductive ☐

Which of the following combinations of components represents an impedance of $110 + j 314 \Omega$ at a frequency of 100 Hz

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- a. An inductor of 50 mH in series with a capacitor of 5 μ F ☐
- b. A resistor of 314 Ω in series with an inductor of 5 mH ☐
- c. A resistor of 110 Ω in series with an inductor of 500 mH ☒
- d. A resistor of 110 Ω in series with a capacitor of 5 μ F ☐

Consider the circuit shown below where $V = 2\text{ V}$. The value of the voltage V_{ab} =



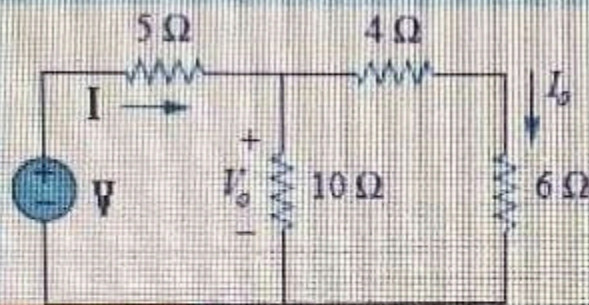
Select one:

☐ a. 18 V

☒ b. 10.5 V

☐ c. 4 V

Consider the circuit shown below. When the voltage $V_o = 4V$, then the value of I is



Select one:

- ☒ a. 0.8 A
- ☐ b. 0.4 A
- ☐ c. 1.2 A